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Credit - Drishti Ias

Tackling Adolescent Malnutrition in India

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Source: TH

Why in News?

The **National Family Health Survey-6 (NFHS-6) (2023-24)** has highlighted the growing challenge of **adolescent malnutrition in India** , with a sharp rise in **obesity , high blood sugar, and lifestyle-related diseases** , emphasizing the need for school-based nutrition and physical activity interventions.

What is the Status of Adolescent Malnutrition in India?

- **Double Burden of Malnutrition:** India continues to face persistent **undernutrition** alongside rapidly rising **overweight and obesity** , particularly among adolescents.
- **Rising Lifestyle Diseases (NFHS-6, 2023–24):** Obesity increased to **30.7% among women** and **27.3% among men** , while the prevalence of **high blood sugar** rose to **17.8% among women** and **20.9% among men** , indicating worsening metabolic health.
  - These issues are no longer just an "urban problem." Sedentary behaviors, stress, and processed diets are driving a steep increase in obesity and metabolic risks in rural adolescents as well.
- **Thin-Fat Phenotype:** Many adolescents appear lean but have **excess visceral fat and high metabolic risk** , making them vulnerable to **early-onset Type-2 diabetes, hypertension, and cardiovascular diseases** .
  - According to the **Comprehensive National Nutrition Survey (CNNS) (2019)** , **27.4% of adolescents are stunted** , while many remain metabolically unhealthy despite appearing lean. Additionally, **35% of stunted children under five** already have **adult-level triglycerides** , increasing their risk of **early-onset diabetes and cardiovascular diseases** .
- **Poor Dietary Patterns:** Adolescents consume **cereal-heavy diets** with inadequate **proteins, fruits, vegetables, and dairy products** , while consumption of **High Fat, Sugar and Salt (HFSS) foods** and **Ultra-Processed Foods (UPFs)** is rising rapidly, with **UPF consumption increasing by over 13.7% annually** .
  - Studies reveal a heavy reliance on cereal-heavy diets, while proteins, dairy, fruits, and green leafy vegetables fall far below the recommendations set by the **Dietary Guidelines for Indians 2024** (which advise that half the plate by volume should be fruits and vegetables).
- **Growing Future Burden:** A **2025 Lancet** study projects that by **2050** , **21.8 crore men** and **23.1**

crore women in India will be overweight, with the steepest increase among the **15–24 years** age group.

- **Physical Inactivity:** Physical inactivity has become a **major public health threat** affecting both urban and rural adolescents.
  - Excessive screen time and sedentary lifestyles are linked to poor dietary habits, increasing the risk of **obesity and cardiovascular diseases** .

How can Schools Address Adolescent Malnutrition in India?

- **Skill-Based Learning:** Schools should equip students with practical skills such as **reading food labels, understanding portion sizes, and identifying misleading junk food marketing** .
  - Led by the **Indian Council of Medical Research-National Institute of Nutrition (ICMR-NIN)** , the ***Let's Fix Our Food*** **(LFOF)** consortium promotes healthier food environments through nutrition literacy and food label reading kits.
  - In a landmark 2025 move, the **CBSE has mandated over 24,000 schools to install "Sugar Boards"** to raise awareness about the hidden sugar content in packaged foods and beverages.
  - Schools should use **interactive learning tools** such as the **NCERT–UNESCO "Let's Move Forward" comic book** to improve nutrition literacy and raise awareness about **healthy diets, hidden sugars, and physical well-being** .
- **Healthy School Food Environment:** Schools should create UPF- and HFSS-free campuses by implementing **FSSAI 's Eat Right School Initiative** , which promotes safe, healthy, and nutritious food through curricular and co-curricular activities, alongside healthy canteen policies.
  - Schools must enforce **FSSAI's 2020 regulations, which restrict the sale of foods High in Fat, Salt, and Sugar (HFSS)** within 50 meters of school campuses.
  - Platforms like **PM POSHAN** should **promote protein-rich, millet-based balanced meals,** while school gardens and fruit breaks can encourage regular consumption of fresh, seasonal produce and foster healthy eating habits from an early age.
- **Curriculum Overhaul:** Nutrition education must shift from basic biology textbooks to practical, everyday life skills. Understanding macronutrients and micronutrients should be as fundamental as basic mathematics.
  - Schools should make **structured physical education and sports a core part of the daily curriculum** , not optional activities. Regular physical activity can counter rising **sedentary lifestyles and excessive screen time** , which are linked to poor dietary habits and a higher risk of **obesity and other non-communicable diseases** .
- **Institutionalizing Periodic Health Screenings:** Schools must move beyond basic height and weight tracking by integrating with the **Rashtriya Bal Swasthya Karyakram (RBSK)** .
  - By hosting periodic **Adolescent Health and Wellness Days (AHD)** on campus, schools can facilitate routine screening for early metabolic risks, **Body Mass Index (BMI)** deviations, and anaemia, ensuring prompt referrals to local primary health centres.
- **Community and Parental Engagement:** Children often consume what is bought at home. Conducting parental awareness sessions on the dangers of UPFs and the necessity of dietary diversity is essential for sustaining school-taught habits.
- **Taxation and Advertising Regulations:** On a macroeconomic level, the government must support school initiatives by imposing higher taxes on sugary beverages and strictly regulating surrogate advertising of junk food targeting minors.

Conclusion

Realising the vision of a *Fit India* requires transforming schools from centres of learning into hubs of preventive healthcare. Complementing blackboards with Nutrition Boards, UPF-free canteens, and mandatory physical activity can help prevent adolescent malnutrition and lifestyle diseases, ensuring that **India's demographic dividend becomes a national asset** rather than a future healthcare burden.

Drishti Mains Question:

"Preventive healthcare must begin in classrooms rather than hospitals." Examine this statement in the context of adolescent nutrition and lifestyle diseases in India.

Frequently Asked Questions (FAQs)

1. What is meant by the 'double burden of malnutrition' in India?
India simultaneously faces **persistent undernutrition** and rapidly rising **overweight, obesity, and lifestyle diseases** , particularly among adolescents.

2. What is the significance of the FSSAI's Eat Right School Initiative?
The initiative promotes **safe, healthy, and nutritious food environments** in schools through nutrition education, healthy canteens, and curricular and co-curricular activities.

3. What are CBSE's 'Sugar Boards' and why are they important?
CBSE has directed over **24,000 affiliated schools** to install **Sugar Boards** to educate students about the hidden sugar content in packaged foods and beverages, encouraging healthier food choices.

4. How can schools help tackle adolescent malnutrition?
Schools can promote **nutrition literacy, UPF-free campuses, balanced meals under PM POSHAN, mandatory physical activity, and periodic health screenings through RBSK** , fostering lifelong healthy habits.

5. Why is the 'thin-fat' phenotype a growing public health concern?
Individuals with the **thin-fat phenotype** appear lean but carry excess visceral fat, increasing their risk of **Type-2 diabetes, hypertension, and cardiovascular diseases** despite having a normal body weight.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/qS7febpKHKI]**

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims:

Q. Which of the following is/are the indicators/ indicators used by IFPRI to compute the Global Hunger Index Report? (2016)

1. Undernourishment
2. Child stunting
3. Child mortality

Select the correct answer using the code given below:

(a) 1 only

(b) 2 and 3 only

(c) 1, 2 and 3

(d) 1 and 3 only

Ans: C

Mains:

Q. How far do you agree with the view that the focus on lack of availability of food as the main cause of hunger takes the attention away from ineffective human development policies in India? **(2018)**

India’s First 3D-Printed Artificial Reefs

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Source: IE

Why in News?

Tamil Nadu is set to deploy **India’s first 3D-printed artificial reef modules** under the second phase of the **Pradhan Mantri Matsya Sampada Yojana** to support marine habitat restoration.

What are the Key Facts About India's First 3D-Printed Artificial Reefs?

- **About:** The pilot deployment is scheduled off the **Ramanathapuram coast** and will test six newly developed reef designs in coastal waters.
    - Scientists will assess the durability and ecological performance of the modules under local coastal conditions before considering larger-scale deployment.
- **Collaboration:** The project is being carried out in collaboration with the Visakhapatnam Regional Centre of **ICAR-Central Marine Fisheries Research Institute** . Each module weighs about 1 tonne and has been developed by Chennai-based startup Tvasta, an IIT Madras-incubated company.
- **Design:** Unlike conventional reinforced concrete reefs, these modules are made without iron reinforcement and have c **omplex 3D-printed structures with crevices, folds and attachment surfaces.**
    - The design aims to **increase porosity, surface area and habitat suitability** for corals, sponges and other reef-associated marine organisms.
- **Impact:** The initiative, funded jointly by the **Centre and State in a 60:40 ratio, is part of a wider artificial reef project across 213 sites in Tamil Nadu** and is expected to benefit traditional fishing communities through **improved fish stocks and healthier marine ecosystems.**

What is a 3D Printed Artificial Reef?

- **About:** 3D-printed artificial reefs are **human-made underwater structures** designed to **restore marine habitats** and **support biodiversity** where natural reef systems are weak or degraded.
- **Technology:** They are created using **advanced 3D-printing methods** that allow more **intricate, customised and species-specific reef designs** than conventional concrete structures.
- **Design:** These reefs can include varied **cavity sizes, folds, textures and complex surfaces that provide shelter,** attachment spaces and breeding grounds for marine organisms.
- **Biodiversity:** Their structure supports colonisation by **corals, sponges, algae, crustaceans, fish and other reef-associated species,** helping build a new marine ecosystem.
- **Fisheries:** By creating additional habitats and nursery grounds, 3D-printed reefs can **increase fish biomass and support sustainable fisheries management** .
- **Advantages:** They offer **flexibility in design, faster modification, site-specific material choices** and the ability to target particular marine species or ecological needs.
- **Concerns:** Their success depends on **durability, non-toxic materials, proper placement and long-term ecological monitoring** , as artificial reefs do not always meet their intended goals.
- **Significance:** 3D-printed artificial reefs represent an i **nnovative restoration tool that can strengthen marine biodiversity, support coastal communities** and reduce pressure on natural coral reefs.

Pradhan Mantri Matsya Sampada Yojana (PMMSY)

- **About:** Pradhan Mantri Matsya Sampada Yojana (PMMSY) was launched on 10th September 2020 to bring a **"Blue Revolution"** by making India's fisheries sector sustainable, productive and inclusive.
- **Objective:** The scheme aims to **increase fish production** , improve productivity, modernise the value chain, strengthen post-harvest infrastructure and improve quality management.
- **Funding:** PMMSY was approved with a total investment of ₹20,050 crore for 2020-21 to 2024-25 and has now been extended up to 2025-26.
- **Structure:** It has two components — the **Central Sector Scheme** , fully funded by the Centre, and the Centrally Sponsored Scheme, implemented by States with partial central support.
- **Focus Areas:** The scheme supports **production enhancement, fisheries infrastructure,** post-harvest management, women's participation, climate-resilient coastal villages, biofloc technology and digital platforms.
- **Achievements:** Under PMMSY, India's fish production rose to 195 lakh tonnes in 2024-25, exports increased, 58 lakh livelihoods were created, and 99,018 women were empowered.

Frequently Asked Questions (FAQs)

1. What are 3D-printed artificial reefs?
3D-printed artificial reefs are human-made underwater structures created using 3D-printing technology to provide habitats for marine organisms.

2. Where will India's first 3D-printed artificial reef modules be deployed?
They will be deployed off the Ramanathapuram coast in Tamil Nadu.

3. Which institutions developed the reef modules?
The modules were developed by Chennai-based startup Tvasta in collaboration with ICAR-CMFRI.

4. Under which scheme is the project being implemented?
The project is being implemented under the second phase of the Pradhan Mantri Matsya Sampada Yojana.

5. How are 3D-printed reefs different from conventional artificial reefs?
They have complex geometries, crevices, folds and attachment surfaces, offering better habitat spaces for marine organisms.

[Watch Video on YouTube: ▶ https://www.youtube.com/embed/o1nBU30EbXw]

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims

Q. Which of the following have coral reefs? (2014)

1. Andaman and Nicobar Islands
2. Gulf of Kachchh
3. Gulf of Mannar
4. Sunderbans

Select the correct answer using the code given below:

(a) 1, 2 and 3 only
(b) 2 and 4 only
(c) 1 and 3 only
(d) 1, 2, 3 and 4

Ans: (a)

Mains

Q. Discuss the causes of depletion of mangroves and explain their importance in maintaining coastal ecology. **(2019)**

Q. What are the consequences of spreading of ‘Dead Zones’ on marine ecosystems? **(2018)**

Nadaprabhu Kempe Gowda

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Source: PIB

The **Vice-President of India** , paid tribute to **Nadaprabhu Sri Kempe Gowda** on his **517th Birth Anniversary (27th June 2026)** in **Bengaluru** , highlighting his enduring legacy in **urban planning, sustainable development, inclusive governance, and environmental conservation** .

- **Historical Context:** Kempe Gowda (1510 – 1569) descendant of the Morasu Gowda lineage, served as a prominent ruler and chieftain under the **Vijayanagara Empire** .
- **Foundation of Bengaluru:** After receiving permission from **Emperor Achyuta Deva Raya (reigned 1529–42)** , **Kempe Gowda** founded **Bengaluru** in **1537** by constructing the **Bengaluru Fort** and establishing **Bengaluru Pete** (the central market area). He also shifted his capital from **Yelahanka** , laying the foundation for the modern city.
  - He envisioned a city where **farmers, traders, artisans, scholars, and people of different faiths** could live and prosper together, laying the foundation for Bengaluru's identity as **"Mini Bharat."**
- **Progressive Social Reforms:** He is historically celebrated for abolishing regressive societal norms, most notably **prohibiting the amputation of the last two fingers** of the left hand of unmarried women during **“Bandi Devaru”** , a prevailing custom among the **Morasu Vokkaligas** .
- **Infrastructure and Cultural Patronage:** Beyond urban planning, he was a great **patron of art and learning** , significantly contributing to public welfare by building numerous **temples and water reservoirs** to ensure drought resilience and cultural flourishing.
- **Legacy** : The **108-foot Statue of Prosperity** in Bengaluru honours Kempe Gowda. It is recognized by the World Book of Records as the **world's tallest bronze statue of a city's founder.**

Read more: Nadaprabhu Kempegowda